

Methods for Protecting Earth against an Asteroid Strike

Astronomers following asteroid activity in space estimate there is a very small chance an object large enough to destroy a whole city could strike Earth in 2032.

But space agency officials say even if such an asteroid keeps heading on a path toward Earth, the world is now much better-equipped to defend itself against such a threat.

The American space agency NASA recently estimated there was a 3.1 percent chance that asteroid 2024 YR4 would hit Earth on December 22, 2032. That is the highest probability predicted for such a large space rock in modern times.

Richard Moissl is head of the European Space Agency's (ESA) planetary defense office. While recognizing the risk the asteroid could present, he told the French news agency AFP people should not panic over such predictions.

Astronomers have noted that the more data they gather, the odds of a direct asteroid hit are expected to keep rising over time. However, scientists say at a certain point the odds will likely drop down to zero.

Moissl said he thinks it is important to remember that even in the unlikely event the probability keeps rising to 100 percent, the world is "not defenseless."

Here are some methods currently being considered as defensive measures to keep humanity safe in case there is a real threat.

Send a spacecraft to hit it

Only one planetary defense method has been tried against an asteroid. In 2022, NASA's Double Asteroid Redirection Test (DART) sent a spacecraft into the 160-meter-wide Dimorphos asteroid. The effort successfully changed the asteroid's orbit around a larger space rock.

Bruce Betts is chief scientist for the nonprofit Planetary Society. He told AFP that space agencies could hit the 2024 YR4 asteroid with several spacecrafts, observing how each one changed the path.

The asteroid discovered in December is estimated to be 40-90 meters wide -- about half the size of Dimorphos.

"You have to take care not to overdo it," Moissl warned. He said this is because if a spacecraft only partly destroys an asteroid, it could send smaller pieces of the space rock heading toward Earth.

Non-contact methods

A separate idea would involve sending a large spacecraft to fly alongside a threatening asteroid. The spacecraft would not touch the asteroid, but would use its gravitational force to pull it away from Earth.

Moissl said another non-contact plan would put a spacecraft near the asteroid to eject a continuous flow of atoms to push the asteroid off course.

Scientists have also considered painting one side of the asteroid white. They believe this could increase the light the object reflects to make it slowly change course.

Contact methods

One idea is to use a nuclear weapon against a threatening asteroid. In laboratory tests, researchers found that X-rays from a nuclear blast could move a rock. But this is considered more of a plan for kilometers-wide asteroids like the one that killed off the dinosaurs. And this method also carries the risk that a nuclear explosion could send additional pieces of the asteroid falling toward Earth.

A similar method – but one considered less dangerous – would involve shooting laser beams from a spacecraft to destroy the side of an asteroid in an effort to push it away from Earth.

If all else fails

Moissl said that if all else fails, at least the world will have a good idea where a threatening asteroid would strike. Since astronomers believe most asteroids would at most threaten to destroy one city, efforts could be organized to get people out of an area before a strike.

"Seven-and-a-half years is a long time to prepare," Moissl added. He also noted that even with the rising odds involving 2024 YR4, there is still about a 97 percent chance the asteroid will miss Earth.

I'm Jill Robbins.

Daniel Lawler with Issam Ahmed reported this story for Agence France-Presse. Jill Robbins adapted it for Learning English.

Vocabulary

asteroid - น. ดาวเคราะห์น้อย

reflect – v. สะท้อน (แสง/เสียง)

Reading Comprehension Quiz

1. What is the primary concern astronomers have identified regarding asteroid activity?

- A) The possibility of a city-destroying object striking Earth in 2032.
- B) The rapid increase in the number of asteroids approaching Earth.
- C) The lack of technology to track asteroid movements effectively.
- D) The risk of asteroids hitting other celestial bodies.

2. What is the general assessment of Earth's defense capabilities against an asteroid threat, according to space agency officials?

- A) Earth is entirely defenseless against such a threat.
- B) The world is now much better-equipped to defend itself.
- C) Defense capabilities are still in the early stages of development.
- D) Current defenses are only effective against very small asteroids.

3. What is the specific asteroid mentioned by NASA with the highest predicted probability of hitting Earth in modern times, and what is that probability?

- A) Asteroid 2032 YR4 with a 3.1% chance.
- B) Asteroid 2024 YR4 with a 3.1% chance.
- C) Asteroid 2024 YR4 with a 97% chance.
- D) Asteroid Dimorphos with a 3.1% chance.

4. What advice does Richard Moissl, head of ESA's planetary defense office, give regarding predictions about asteroid impacts?

- A) People should immediately prepare for evacuation.
- B) People should not panic over such predictions.
- C) Space agencies should increase their budget for defense.
- D) The risk is negligible and can be ignored.

5. How do astronomers anticipate the odds of a direct asteroid hit will change as more data is gathered?

- A) The odds will steadily decrease to zero from the start.
- B) The odds are expected to keep rising over time, then likely drop to zero.
- C) The odds will remain constant until the impact date.
- D) The odds will fluctuate unpredictably without a clear pattern.

6. Even if the probability of an asteroid hit rises to 100 percent, what is Moissl's reassurance?

- A) Humanity can simply relocate to another planet.

- B) The world is "not defenseless."
- C) The impact would be minor and cause no harm.
- D) Such an event is scientifically impossible.

7. What was the main objective and outcome of NASA's DART mission in 2022?

- A) To observe the composition of the Dimorphos asteroid without contact.
- B) To send a spacecraft to hit an asteroid and successfully change its orbit.
- C) To destroy the Dimorphos asteroid into smaller, harmless pieces.
- D) To test a new propulsion system for deep space travel.

8. According to Bruce Betts, how could 2024 YR4 be addressed, specifically?

- A) By observing its path without intervention.
- B) By using a single, powerful spacecraft to deflect it.
- C) By hitting it with several spacecrafts and observing the path changes.
- D) By deploying a large net to catch the asteroid.

9. What is the estimated size of asteroid 2024 YR4 compared to Dimorphos?

- A) It is twice the size of Dimorphos.
- B) It is about half the size of Dimorphos.
- C) It is roughly the same size as Dimorphos.
- D) Its size is unknown compared to Dimorphos.

10. Why does Richard Moissl warn against "overdoing it" when trying to destroy an asteroid with a spacecraft?

- A) It could waste valuable resources unnecessarily.
- B) It might cause the asteroid to accelerate towards Earth.
- C) Partial destruction could send smaller pieces towards Earth.
- D) It could damage the spacecraft permanently.

11. Which non-contact method involves using a spacecraft's own force to alter an asteroid's path?

- A) Detonating an explosive device nearby.
- B) Using its gravitational force to pull the asteroid away.
- C) Firing continuous laser beams at its surface.
- D) Painting one side of the asteroid a different color.

12. Besides gravitational pull, what other non-contact method does Moissl suggest to push an asteroid off course?

- A) Creating a magnetic field around the asteroid.
- B) Ejecting a continuous flow of atoms near the asteroid.
- C) Using sound waves to vibrate the asteroid.
- D) Building a giant solar sail to capture the asteroid.

13. What unique non-contact method involves altering the asteroid's surface to change its trajectory?

- A) Covering the asteroid with reflective foil.
- B) Drilling holes into the asteroid to vent gas.
- C) Painting one side of the asteroid white to increase light reflection.
- D) Attaching small rockets to the asteroid's surface.

14. Which contact method is primarily considered for kilometers-wide asteroids like the one that killed the dinosaurs, and what is its associated risk?

- A) Sending a powerful spacecraft to ram it, with the risk of the spacecraft breaking apart.
- B) Using laser beams to ablate its surface, with the risk of creating a dust cloud.
- C) Using a nuclear weapon, with the risk of sending additional pieces falling towards Earth.
- D) Attaching rockets to steer it, with the risk of the rockets failing to ignite.

15. What is a contact method considered less dangerous than a nuclear weapon, involving laser beams?

- A) Shooting laser beams from Earth to vaporize the asteroid.
- B) Deploying a laser-guided missile from a spacecraft.
- C) Shooting laser beams from a spacecraft to destroy the side of an asteroid to push it away.
- D) Using lasers to create a deflecting force field around the asteroid.